-Natural Language Processing (NLP) consists of various stages, first speech analysis (transforming speech into text), followed by morphological and lexical analysis, which is then parsed, the text is subjected to contextual reasoning.

-NLP can be categorized in multiple stages, these are: Phonology (study of linguistic sound), morphology (study of meaningful components of words), syntax (study of structural relationships between words), semantics (the study of meaning), pragmatics (the study of how language is used to accomplish goals), and discourse (the study of larger linguistic units).

-The process of speech to text (STT) can be done using open-source toolkit such as Kaldi.

-NLP is challenging due to the huge amount of context, different meaning sentences can mean and complex syntax. It relies on comprehension and not precise closed artificial structures.

-Research trends in NLP moved from rigid rule-based approaches to statistical approaches in the 1990s and now focus on powerful and flexible quantitative approaches such as deep learning or end to end multi-task learning.

-Performance improvements are constantly made, various models are made to achieve state of the art (SOTA) such as the BERT, ELMo or Self Attention.

-Changes in model structure are constantly made through human input, improving the model.

-Amazon SageMaker, Scale Ai, Basic AI are data labelling platforms.

-Testworks offers one stop solution, like smart solution for data collection and data labelling solution with automation.

-The goal of lexical analysis is to convert a sequence of characters into a sequence of tokens. In NLP morpheme is a basic unit and in text mining word is commonly used as a basic unit for analysis.

-The process of lexical analysis is tokenizing, part of speech (POS) tagging and additional analysis such as named entity (NER), noun phrase recognition, sentence split and chunking.

-Sentence splitting is a very important part of NLP, but very challenging because sentence units like “.” doesn’t always mean the end of a sentence. This is why sentence boundaries are very important.

-In lexical analysis, text split into basic units is called tokens. It can include word tokens, number tokens and space tokens.

-Morphological analysis deals with the morphological variation in words, for example car and cars or give gave and given. The goal is to normalize the word into the same form. The techniques are stemming and lemmatization.

-Stemming is the process of reducing a word to its base form, using rule-based algorithms that remove suffixes.

-Lemmatization is the process of deriving the base form of a word from one of its inflected forms. The process is more complex than stemming. It identifies the root form of the word.